



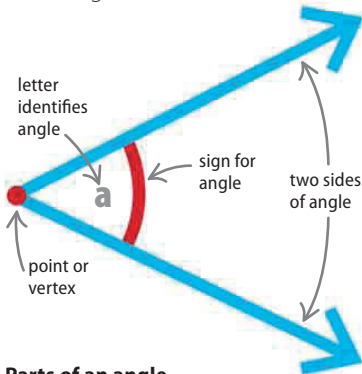
Angles

AN ANGLE IS A FIGURE FORMED BY TWO RAYS THAT SHARE A COMMON ENDPOINT CALLED THE VERTEX.

Angles show the amount two lines “turn” as they extend in different directions away from the vertex. This turn is measured in degrees, represented by the symbol $^{\circ}$.

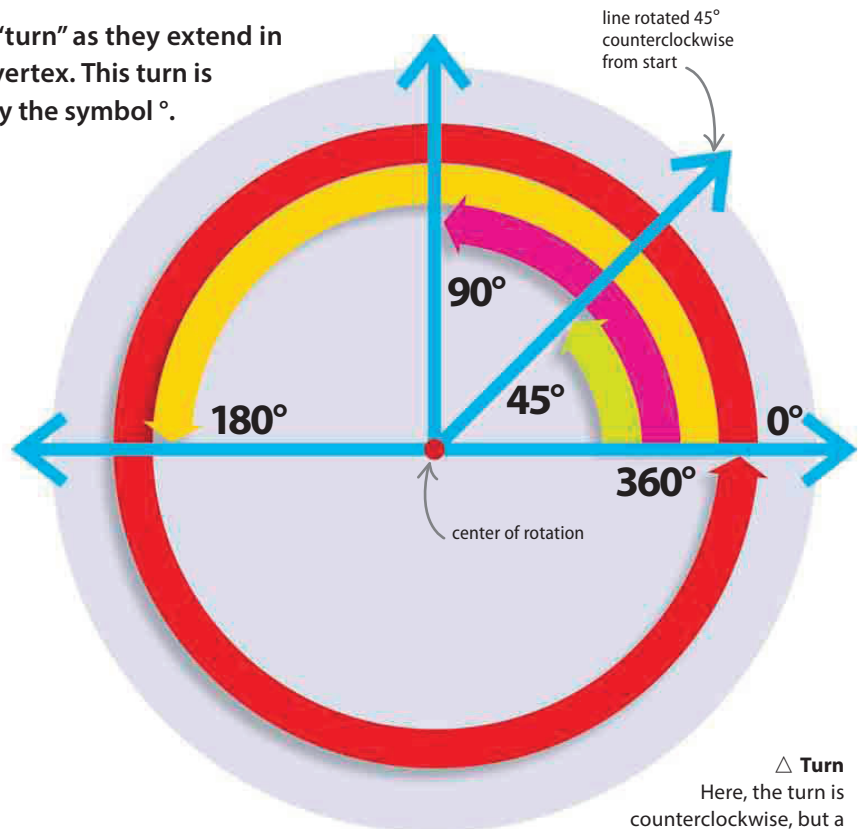
Measuring angles

The size of an angle depends on the amount of turn. A whole turn, making one rotation around a circle, is 360° . All other angles are less than 360° .



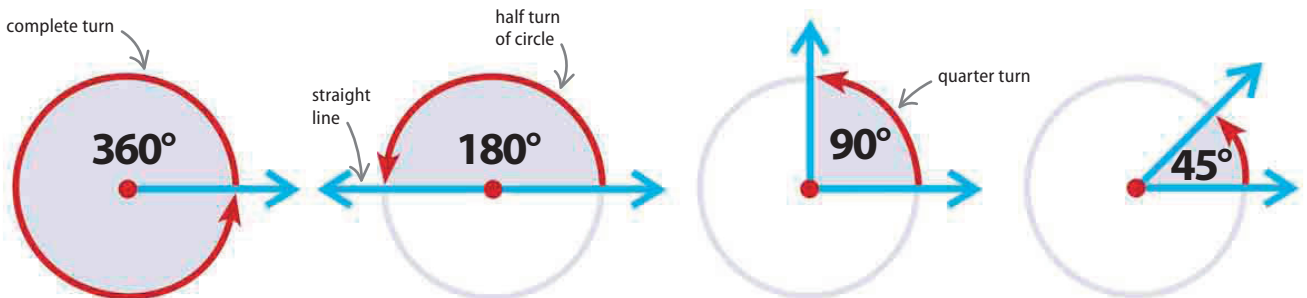
△ Parts of an angle

The space between these two rays is the angle. An angle can be named with a letter, its value in degrees, or the symbol $\angle$.



△ Turn

Here, the turn is counterclockwise, but a turn can also be clockwise.



△ Whole turn

An angle that is a whole turn is 360° . Such a rotation brings both sides of the angle back to the starting point.

△ Half turn

An angle that is a half turn is 180° . Its two sides form a straight line. The angle is also known as a straight angle.

△ Quarter turn

An angle that is a quarter turn is 90° . Its two sides are perpendicular (L-shaped). It is also known as a right angle.

△ Eighth turn

An angle that is one eighth of a whole turn is 45° . It is half of a right angle, and eight of these angles are a whole turn.

SEE ALSO

◀ 82–83 Tools in geometry

Straight lines 86–87 ▶

Bearings 108–109 ▶